2025-Winter Paper 4 – Recent Advances

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Q1. Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer?

Type: LAQ (25 marks)

Introduction

- After RC (radical cystectomy) alone, nearly half of MIBC (muscle-invasive bladder cancer) patients develop metastases from occult micrometastases.
- Cure improves with **systemic therapy before surgery** and a **meticulous lymph node dissection** (accurate staging, possible therapeutic value).
- UTUC (upper tract urothelial carcinoma) behaves alike, but RNU (radical nephroureterectomy) removes a kidney, changing chemotherapy timing.

A. Neoadjuvant therapy in MIBC

Rationale

- Treats micrometastases early; better tolerated before surgery, so more complete it.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

Selection: cT2–T4a N0 M0, cisplatin-eligible. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, hearing loss grade ≥2, neuropathy grade ≥2, NYHA (New York Heart Association) class III–IV failure.

Trial	Regimen	Result
SWOG 8710 (Southwest Oncology Group)	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC vs RC	pT0 38% vs 15%; 5-year OS (overall survival) 57% vs 43%, **P = 0.06**
BA06 30894 (EORTC/MRC, European Organisation for Research and Treatment of Cancer/Medical Research Council)	CMV (cisplatin, methotrexate, vinblastine) ×3	16% relative reduction in mortality; 10-year OS 36% vs 30%
ABC (Advanced Bladder Cancer) meta-analysis	Cisplatin combinations	OS HR (hazard ratio) 0.86; **5% absolute benefit** at 5 years

Trial	Regimen	Result
VESPER	ddMVAC (dose-dense MVAC) ×6 vs GC (gemcitabine–cisplatin) ×4	Organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity
NIAGARA	Durvalumab + GC, then adjuvant durvalumab, vs GC	Better event-free survival (HR 0.68) and OS (HR 0.75)

- Single-agent cisplatin, carboplatin and neoadjuvant radiotherapy give **no survival benefit**.
- **Guidelines (EAU, European Association of Urology; AUA, American Urological Association; NCCN, National Comprehensive Cancer Network):** cisplatin-based NAC (neoadjuvant chemotherapy) for all eligible patients; ddMVAC 3–6 cycles or GC 4 cycles. NIAGARA has since added perioperative durvalumab + GC (the chapter predates it and calls immunotherapy unproven).
- **Cisplatin-ineligible:** proceed to RC; neoadjuvant pembrolizumab (PURE-01) and atezolizumab (ABACUS) are investigational.
- **Adjuvant** (pT3–4 or N+): cisplatin chemotherapy if not given before; **nivolumab** (CheckMate 274: disease-free survival 20.8 vs 10.8 months).

B. Lymphadenectomy in MIBC

- **Primary drainage:** obturator, internal and external iliac, presacral. **Secondary:** common iliac, para-aortic, paracaval.
- Node positivity: pT2 8–26%, pT3 33–50%, pT4 36–80%.

Template	Extent
Standard	Bilateral external and internal iliac and obturator nodes; genitofemoral nerve laterally, node of Cloquet caudally, up to the common iliac bifurcation
Extended	Adds common iliac and presacral nodes, up to the aortic bifurcation
Super-extended	Up to the inferior mesenteric artery

- **Evidence:** German phase III (limited vs extended): no survival benefit. **SWOG S1011** (most had NAC): median nodes 24 vs 39, **no difference** in disease-free survival (HR 1.10) or OS; more grade ≥3 events. So **standard bilateral PLND (pelvic lymph node dissection)** is the standard of care.
- **Node count:** fewer than 10 nodes doubled the risk of death (SWOG 8710 reanalysis). **Also know:** lymph node density, extranodal extension (prognostic).
- Unexpected nodes at RC: frozen section, proceed if resectable within the pelvis. Clinically node-positive: chemotherapy first, then RC in responders.

C. Neoadjuvant and adjuvant therapy in UTUC

- **Why neoadjuvant:** CKD (chronic kidney disease) is common and GFR falls after RNU, losing cisplatin eligibility. Give platinum **while both kidneys work**.
- Biopsy under-stages, so select by high-risk features (high grade, invasion on imaging, hydronephrosis).
- **EA8141** (phase II): accelerated MVAC, or gemcitabine–carboplatin if CrCl (creatinine clearance) 30–49; complete response 14%, below ypT2N0 60%.
- **AUA/SUO (Society of Urologic Oncology):** neoadjuvant platinum is a standard option for high-risk UTUC, especially if postoperative CrCl will fall below 50.
- **Adjuvant: POUT** (phase III, pT2–T4 or N+ after RNU): gemcitabine–platinum vs surveillance; disease-free survival HR 0.45, 3-year 71% vs 46%. Weaker with carboplatin, favouring neoadjuvant cisplatin. Adjuvant nivolumab (CheckMate 274) also included UTUC.

D. Lymphadenectomy in UTUC

- Node positivity 12–25%. Lymph drains **right to left and top to bottom**, so the template depends on side and level.

Primary site	Right	Left
Renal pelvis, upper two-thirds ureter	Hilar, paracaval, retrocaval, **interaortocaval**	Hilar, para-aortic
Lower third ureter	Ipsilateral common, external, internal iliac, obturator, presacral	Same

- Value: pN0 better than pNx; ≥8 nodes improved disease-specific survival above pT1. Indicated for high-grade or invasive disease; robotic yield equals open. Risk: chylous leak.

Diagram to draw: standard vs extended pelvic templates, and right vs left UTUC templates.

> **Key points to remember:**
>
> - Cisplatin combinations only: ABC 5% absolute 5-year survival gain; SWOG 8710 P = 0.06.
> - VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
> - SWOG S1011: extended PLND gives no survival benefit over standard.
> - UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.

Q2. Describe Basic Energy Modalities in Urological surgery?

Type: LAQ (25 marks)

Introduction

- Every modality delivers energy that tissue absorbs and releases as **heat, mechanical stress or cavitation**. Frequency sets penetration; absorption sets the effect.
- Tissue effects of heat: about 60 °C coagulates protein; 100 °C vaporises cell water; higher temperatures carbonise.

Classification

Energy	Examples
Electrical	Monopolar, bipolar, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration)
Electromagnetic	Lasers, MWA (microwave ablation)
Acoustic	Ultrasonic dissector, SWL (shock wave lithotripsy), HIFU (high-intensity focused ultrasound), histotripsy
Direct contact	Cryoablation, water vapour, waterjet (Aquablation)

1. Electrosurgery

- **Physics:** alternating current at about **500 kHz** (radiofrequency). Mains current (50 Hz) depolarises nerve and muscle; at 500 kHz ions only oscillate and produce heat.
- **Ohm's law** ($V = I \times R$): tissue is the resistance. **Current density** decides the effect: a small tip concentrates current and heats; a large pad disperses it.
- **Monopolar:** generator → active electrode → patient → dispersive pad → generator. Partial pad contact causes burns.
- **Bipolar:** current passes only between two jaws; less power, less spread, no pad.
- **Electrocautery** (heated wire) differs: no current enters the patient.

Waveform	Character	Effect
Cut	Continuous, low voltage; 50–80 W	Rapid boiling, clean incision, little char
Coagulation	Interrupted (about 6% duty cycle), high voltage; 30–50 W	Desiccation; spread up to 3 mm
Fulguration (spray)	High voltage arcs across air	Non-contact surface haemostasis
Blend	Cut with interruptions	Cutting plus haemostasis

- **Vessel sealers:** bipolar current plus jaw compression fuses collagen and elastin; a feedback generator stops when impedance rises. Seal vessels up to 7 mm in one application, with little spread.
- **Endoscopic:** monopolar TURP (transurethral resection of prostate) needs non-conducting **1.5% glycine**, risking hyponatraemia (TUR syndrome); **bipolar** uses saline, with comparable efficacy and better safety.
- **Obturator reflex:** lateral-wall TURBT (transurethral resection of bladder tumour) stimulates the nerve → adduction → perforation. Prevent with paralysis, obturator block, partial filling, short bursts, bipolar.
- **Safety:** pad burns; insulation failure; **direct coupling** (touching metal); **capacitive coupling** (current induced through intact insulation); fire with oxygen; inadvertent activation. Use AEM (active electrode monitoring), lowest power, bipolar near pacemakers.

2. Ultrasonic energy

- **Ultrasonic dissector:** blade vibrates at about 55.5 kHz; friction and cavitation denature protein. No current through the patient; little lateral spread, but the blade stays hot.
- Intracorporeal ultrasonic lithotripter: vibrating probe, suction of fragments.

3. Lasers

- **Principle:** pumped gain medium in a mirrored cavity → stimulated emission → **monochromatic, coherent, collimated** beam.
- **Effect** depends on wavelength (chromophore: water, haemoglobin) and pulse duration.

Laser	nm	Penetration (mm)	Use
KTP/LBO (potassium-titanyl-phosphate/lithium triborate)	532	0.8	Prostate photovaporisation (haemoglobin)
Nd:YAG (neodymium:yttrium-aluminium-garnet)	1064	10	Deep coagulation
TFL (thulium fibre laser)	1940	0.08	Stones, enucleation
Tm:YAG (thulium:YAG)	2000	0.2	Vaporesection, enucleation
Ho:YAG (holmium:YAG)	2120	0.31	Stones, HoLEP (holmium laser enucleation of prostate), strictures

- **Mechanisms:** photothermal (long pulses), photomechanical (short pulses, shock waves), photochemical (PDT, photodynamic therapy).
- **Penetration predicts behaviour:** deep (Nd:YAG) coagulates widely with delayed sloughing; shallow (thulium) vaporises precisely with char.
- Safety: eye protection for all; irrigant heating; fibre damage.

4. Ablative energies

Modality	Mechanism	Key points
RFA	375–900 kHz current; ionic agitation	Target 60 °C; char at 150 °C; heat sink
MWA	915 MHz or 2.45 GHz; dielectric hysteresis	Hotter, larger zones; less heat sink
Cryoablation	Argon expansion (Joule–Thomson)	Below −40 °C; **two freeze–thaw cycles**; ice ball visible
HIFU	Focused ultrasound absorbed → necrosis	1500–10,000 W/cm²; rice-grain lesions; transrectal prostate
IRE	High-voltage microsecond pulses → permanent membrane pores	Non-thermal; spares vessels and ducts; focal prostate

- Cryo mechanism: extracellular ice → cell dehydration → intracellular ice → membrane rupture on thawing; ischaemia and apoptosis follow.
- IRE mechanism: electric pulses open pores in cell membranes that do not reseal, so cells die by apoptosis without heat; collagen scaffolds survive. It needs general anaesthesia with muscle paralysis and ECG (electrocardiogram)-synchronised pulses.
- **Also know:** histotripsy (cavitation liquefaction, investigational); water vapour (Rezum, 103 °C, convective, no char); Aquablation (non-thermal saline jet); chemohyperthermia (42 °C) and EMDA for mitomycin.

5. Lithotripsy energies

- **SWL:** electrohydraulic, electromagnetic or piezoelectric sources. Stones break by **spallation, shear and cavitation**. Slower rates (about 1 Hz) work better and injure less.

- **Intracorporeal:** pneumatic (ballistic), ultrasonic, combined, electrohydraulic, laser. **Also know:** burst wave lithotripsy, ultrasonic propulsion.

Diagram to draw: monopolar vs bipolar circuit, with cut and coagulation waveforms.

> **Key points to remember:**
> - Electrosurgery at about 500 kHz; current density decides heating.
> - Cut continuous low voltage; coagulation interrupted high voltage.
> - Capacitive coupling and insulation failure cause unseen burns.
> - Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink.

Q3a. Immuno modulation in Renal, cell carcinoma

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 2, Q3e)

Assumed meaning: immunomodulation (immunotherapy) in renal cell carcinoma.

Rationale

- RCC (renal cell carcinoma) is immunogenic: rare (<1%) spontaneous regression after nephrectomy, tumour-infiltrating T cells.
- **Checkpoints:** PD-1 (programmed death-1) on T cells binds PD-L1 on tumour and switches T cells off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) damps priming. Blockade **releases the brake** on existing immunity.
- VEGF (vascular endothelial growth factor) suppresses immunity, so TKIs (tyrosine kinase inhibitors) synergise.

Cytokines (non-specific stimulation)

	IFN-α (interferon-alpha)	High-dose IL-2 (interleukin-2)
Response	10–15%	15–20%
Durable complete response	<2%	**7–9%**
Toxicity	Flu-like	**Capillary leak**; intensive care
Role	Obsolete	Selected fit patients

- High-dose IL-2: 600,000–720,000 IU/kg intravenously every 8 hours, maximum 15 doses; over 60% of complete responders stay disease-free.

Checkpoint inhibitors (clear cell)

Trial	Regimen vs comparator	Result
CheckMate 025	Nivolumab vs everolimus, second line	OS (overall survival) 25.0 vs 19.6 months
CheckMate 214	Nivolumab + ipilimumab vs sunitinib	Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9%
KEYNOTE-426 / CheckMate 9ER	Pembrolizumab + axitinib / nivolumab + cabozantinib	Better PFS (progression-free survival) and OS
CLEAR	Pembrolizumab + lenvatinib	PFS 23.9 vs 9.2 months; response 71%

- **First line:** IO/TKI (immuno-oncology plus TKI) for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor only.
- After IO, switch to a VEGF-targeted TKI; IO rechallenge is not recommended.

Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC (including resected metastases): disease-free survival HR 0.63; later OS benefit.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis. Grade 2: hold, prednisolone; grade 3–4: stop, high-dose steroids.

Also know: PD-L1 does not select; sarcomatoid responds well; vaccines and allogeneic transplant failed.

Diagram to draw: T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

Key points to remember:

- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63.
- KEYNOTE-564: adjuvant pembrolizumab.

Q3b. New LASER for stone Treatment

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3c)

Background

- **Ho:YAG (holmium:yttrium-aluminium-garnet), 2120 nm** (often quoted as 2100 nm) is the standard. It is water-absorbed and **photothermal**: long pulses heat the stone until it decomposes.
- Newer lasers aim at less retropulsion, finer dust and faster work.

1. High-power holmium with pulse modulation

- 100–120 W machines allow **dusting** (low energy, high frequency) and popcorning.
- **Moses effect:** the start of each pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the stone.
- **Moses technology** splits each pulse in two: the first opens the channel, the second strikes the stone. Less retropulsion.

2. TFL (thulium fibre laser), 1940 nm

- Diode-pumped silica fibre laser near the **water absorption peak** (higher absorption than holmium); penetration 0.08 mm.
- **Low peak power, long pulses, very high frequency** → finer dust, minimal retropulsion.
- **Thin fibres** (about 150 µm): better irrigation and scope deflection. Compact, quiet.
- **RCTs (randomised controlled trials):** TFL beat **low-power (30 W) holmium** (stone-free rate, time); it **equalled 120 W pulse-modulated holmium** in stone-free rate.

3. Pulsed Tm:YAG (thulium:YAG), 2013 nm

- Solid-state pulsed version of the continuous-wave thulium laser; photothermal lithotripsy and enucleation.

Comparison

	Ho:YAG	TFL
Wavelength	2120 nm	1940 nm
Peak power	High	Low
Retropulsion	More	Less
Fragments	Small	Smaller

Principles of settings

- **Retropulsion rises with pulse energy**, not frequency; it falls with longer pulses.
- Keep the fibre close and perpendicular to the stone.

Safety

- Heat follows total power: use low power, pauses, high flow and an **access sheath**; urothelial injury begins at about 43 °C. Eyewear for all.

Also know: suction sheaths; burst wave lithotripsy (investigational).

Diagram to draw: water absorption curve marking 1940 and 2120 nm.

Key points to remember:

- TFL 1940 nm: low peak power, minimal retropulsion.
- Moses: vapour channel; two-part pulse.
- TFL beats low-power, equals high-power holmium.

Q3c. Combination Intravesical therapy

Type: SAQ (10 marks)

Rationale

- Single agents fail: chemotherapy reduces recurrence only; BCG (bacille Calmette-Guérin) reduces progression but fails in about a third.
- Combinations add **different mechanisms**, overcome resistance, and help during **BCG shortage** and in **BCG-unresponsive** NMIBC (non-muscle-invasive bladder cancer).

1. Sequential chemotherapy (doublets)

Regimen	Method	Results
Gem/Doce (gemcitabine–docetaxel)	Gemcitabine 1 g (60–90 min) → drain → docetaxel 37.5 mg (60–120 min); weekly ×6, then monthly for 2 years	High-grade recurrence-free survival about 65% (1 year), 52% (2 years); 84% avoid cystectomy
Gem/MMC (mitomycin C)	Gemcitabine 1 g → MMC 40 mg	Durable recurrence-free about 38%

- **Why it works:** gemcitabine (antimetabolite) and docetaxel (microtubule stabiliser) act on different cell-cycle phases.
- Oral sodium bicarbonate and antiemetics aid tolerance. Phase III BRIDGE (versus BCG in BCG-naive) is testing it.

2. Chemotherapy with BCG

- Sequential MMC–BCG: **no benefit** over BCG alone.
- **Exception: BCG alternating with EMDA (electromotive drug administration)-MMC** in T1: lower recurrence and progression than BCG alone. A later trial was non-significant.

3. BCG with immunomodulators

- **BCG + interferon-α:** no gain over BCG; allows reduced BCG dose.
- **N-803 (nogapendekin alfa)** = IL-15 (interleukin-15) superagonist activating CD8 T and NK (natural killer) cells. With BCG in BCG-unresponsive CIS (carcinoma in situ): complete response 71%, 45% at 12 months. FDA (Food and Drug Administration)-approved.

4. BCG with checkpoint inhibitors (BCG-naive high risk)

- **CREST** (subcutaneous sasanlimab + BCG) and **POTOMAC** (durvalumab + BCG): improved event-free/disease-free survival over BCG alone.
- **ALBAN** (atezolizumab + BCG): negative. KEYNOTE-676 (pembrolizumab) ongoing.

5. Device-assisted and novel

- **Chemohyperthermia** (MMC at 42 °C: RITE, radiofrequency-induced thermochemotherapeutic effect; HIVEC, hyperthermic intravesical chemotherapy) increases uptake.
- **TAR-200:** gemcitabine-releasing intravesical device; high complete response in BCG-unresponsive CIS (SunRISe-1).
- Cretostimogene + pembrolizumab: about two-thirds complete response at 12 months.

Toxicity

- Additive cystitis; immune-related events with checkpoint inhibitors.

Diagram to draw: Gem/Doce sequence timeline (instil, dwell, drain, instil).

Key points to remember:

- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

Q3d. Penile Inversion Vaginoplasty Technique

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 2, Q3b)

Introduction

- Commonest gender-affirming genital operation for transgender women: a sensate, functional vulva and vagina.
- **Principle:** homologous penile and scrotal tissues on their own blood supply.

Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care 8):** one assessment, **6 months of hormones**.
- **Permanent hair removal** of scrotal graft skin; BMI (body mass index) usually ≤35 kg/m²; stop smoking; fertility preservation.
- **Choose depth:** full-depth vs **vulvoplasty (zero-depth)**: vulva without a canal, chosen for high surgical risk, prior prostatectomy, or no wish for penetrative intercourse.

Tissue transfer

Male structure	Becomes
Glans on dorsal neurovascular bundle	**Clitoris**
Preputial collar	Clitoral hood, labia minora
Spatulated urethra	Vestibule
Inverted penile skin	**Neovaginal lining**
Scrotal full-thickness graft	Proximal lining
Lateral scrotal skin	Labia majora

Steps

```
Lithotomy; tranexamic acid 1 g; catheter
→ harvest, defat, tubularise scrotal graft
→ bilateral orchidectomy
→ deglove penis, keep dartos with skin (blood supply)
→ dorsal neurovascular bundle + glans raised;
   excise ALL erectile tissue
→ clitoroplasty; shorten, spatulate urethra
→ canal dissection between prostate and rectum;
   keep Denonvilliers fascia intact; rectal checks
→ invert skin tube + graft; labiaplasty
→ pack canal
```

- **Why excise all erectile tissue:** residual tissue engorges and narrows the canal.
- **Peritoneal flaps (robotic)** add depth; **intestinal vaginoplasty** when skin is inadequate or for revision.

Postoperative care

- Pack and catheter out about day 5; **lifelong dilation** (several times daily at first, tapering); pelvic floor physiotherapy; continue oestrogen; most achieve orgasm.

Complications

- **Wound dehiscence** (commonest, 12–33%).
- **Rectovaginal fistula** (0.8% primary, 6.3% revision); urethrovaginal fistula (0.8–3.9%).

- Urethral stricture, meatal malposition (spraying); **vaginal stenosis** (poor dilation); prolapse 4–10%.

Diagram to draw: sagittal view of neovagina between prostate and rectum.

> **Key points to remember:**
> - Glans → clitoris; penile skin → canal; scrotal skin → graft.
> - Excise all erectile tissue; preserve Denonvilliers fascia.
> - Lifelong dilation; rectovaginal fistula is worst.

Q3e. Robotic RPLND

Type: SAQ (10 marks)

See 2025-Winter Paper 3, Q3f (full answer).

Key points to remember (recent advances angle):

- Expanding indication: primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy.
- Robotic nerve sparing preserves antegrade ejaculation in ≥90% after primary surgery.
- Serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND (retroperitoneal lymph node dissection).

Q3f. Management of Defecation Disorders resulting into LUTS

Type: SAQ (10 marks)

Assumed meaning: bladder and bowel dysfunction, where constipation or defecation disorders cause LUTS (lower urinary tract symptoms).

Why the bowel causes LUTS

- **Shared space:** a loaded rectum compresses the bladder, reducing capacity and obstructing outflow.
- **Shared nerves:** bladder and bowel afferents converge on the same spinal segments; cross-organ sensitisation causes urgency and detrusor overactivity.
- **Shared muscle:** a pelvic floor that contracts during defecation (**dyssynergic defecation**) also contracts during voiding (**dysfunctional voiding**) → high PVR (post-void residual), UTI (urinary tract infection), reflux.
- Result: urgency, daytime incontinence, enuresis, retention, recurrent UTI.

Evaluation

- Stool frequency, pain, **Bristol chart**; 3-day bladder and bowel diary.
- **Rome IV (child):** at least 2, weekly for 1 month: ≤2 defecations/week; faecal incontinence weekly; retentive posturing; painful or hard stools; large rectal mass; large-diameter stools.
- **Rome IV (adult), for 3 months:** ≥2 of straining, hard stools, incomplete evacuation, blockage, manual manoeuvres (each in >25% of defecations), <3 stools/week.
- Examine abdomen, spine and perineum; exclude organic causes (Hirschsprung disease, hypothyroidism, neurogenic).
- **Also know:** rectal diameter ≥30 mm on ultrasound; no routine X-ray; manometry if refractory.

Management

Rule: treat the bowel first (EAU, European Association of Urology/ESPU, European Society for Paediatric Urology).

1. **Disimpaction:** oral PEG (polyethylene glycol) 3350, 1–1.5 g/kg/day for 3–6 days; enemas if needed.
2. **Maintenance:** PEG 0.4 g/kg/day (range 0.2–0.8), at least 2 months; lactulose alternative; stimulants (senna) for expulsion.
3. **Urotherapy:** toileting after meals (gastrocolic reflex); footstool posture; normal fluid and fibre.
4. **Biofeedback, pelvic floor physiotherapy** for dyssynergia.
5. **Refractory: transanal irrigation** (water via rectal cone; a quarter to a third stop); **ACE (antegrade continence enema, Malone)**; botulinum toxin to the sphincter; sacral or tibial neuromodulation.

- **Adults:** laxatives, prucalopride or linaclotide, biofeedback; rectocele repair; irrigation in neurogenic bowel.
- **Caution:** anticholinergics worsen constipation; LUTS return.

Diagram to draw: pelvic cross-section showing loaded rectum compressing bladder.

> **Key points to remember:**
> - Treat the bowel first.
> - Rome IV: ≥2 criteria for 1 month (child).
> - PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

Rapid Revision Sheet – 2025-Winter Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63.
- KEYNOTE-564: adjuvant pembrolizumab.

Q3b. New LASER for stone Treatment (SAQ)

- TFL 1940 nm: low peak power, minimal retropulsion.
- Moses: vapour channel; two-part pulse.
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